# SAHIL T CHAUDHARY

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## EDUCATION

**Carnegie Mellon University** — *Pittsburgh, PA*

Master of Science in Mechanical Engineering – Research | **GPA:** 4.0/4.0 — *May 2025*

- **Coursework:** Planning and Decision-making, Introduction to Robot Learning, Optimal Control and Reinforcement Learning, Robot Localization and Mapping, Modern Control Theory, Machine Learning, Robot Dynamics and Analysis
- **Scholarship:** Fully funded Master's student
- **Course assistant** for Machine Learning and Artificial Intelligence for Engineers (24787)

**Vellore Institute of Technology** — *Vellore, India*

Bachelor of Technology in Mechanical Engineering | **GPA:** 9.05/10.0 — *May 2022*

## SKILLS

**Programming Languages:** C++, Python, Bash, MATLAB, Julia, Rust
**Tools:** Isaac Lab, Gazebo, Unity, PyTorch, SolidWorks, Fusion 360, Ansys
**Platforms:** ROS, Git, Linux, Docker

## WORK EXPERIENCE

**Juxta Technology, Inc.** — *San Francisco, CA*

*Robotics Engineer, AI* — *September 2025 – Present*

- Developed **learning-based**, **probabilistic**, and **filtering** methods for **drift mitigation** in **neural-inertial navigation and localization**, achieving sub-meter positioning accuracy with **~2 m ATE/RTE**
- Built a **simulation framework** for generating realistic human motion trajectories and **synthetic noisy IMU measurements** for training and evaluation, and incorporated **learning-based data refinement** techniques; current synthetic-only training achieves **~3 m ATE/RTE** on real-world test trajectories

**Biorobotics Lab, CMU Robotics Institute** — *Pittsburgh, PA*

*Graduate Research Assistant (Paid during Summer 2024)* — *August 2023 – May 2025*

- Contributed to the implementation of an **Inverse Kinematics-based optimization** framework for optimal communication node placement
- Spearheaded the development of a **MANET framework** using DDS and ROS to ensure communication fidelity in heterogeneous robot convoys, perform network topology repair and recovery behaviors, and enforce a communication boundary
- **Comms-Aware Planning:** Designed a novel algorithm attaining **99% success rate** to maintain communication among robots over radio, by formulating a modified Max-Min Spanning Tree, and validated the algorithm through extensive hardware testing
- **Heterogeneous Convoy Framework:** Developed a decentralized convoy framework integrating RC cars and quadrupeds like Boston Dynamics' Spot, and algorithms enabling intersection rendezvous and coordinated return for exploration missions
- Enhanced the operational efficiency of the **Local Planner by up to 29%**, through waypoint optimization and trajectory smoothing, reducing unnecessary deceleration between waypoints and improving overall robot speed and motion continuity
- **Payload Redesign:** Engineered a modular, serviceable payload for RC cars and quadruped robots, accomplishing a **7% weight reduction** and lowering the center of gravity while ensuring optimal sensor field-of-view and accessibility, and incorporating sensors such as LiDAR, IMU, cameras, onboard computer, motor controller, and circuit boards

## PUBLICATIONS

**Sahil Chaudhary***, Charles Noren*, Burhanuddin Shirose, Bhaskar Vundurthy, Matthew Travers, "Communication Network Construction Behaviors for Robotic Convoying" — *Published in **GVSETS 2025***

## PROJECTS

**Blind Manipulation** [Website] — *Pittsburgh, PA*

*Search-Based Planning Lab, CMU RI – Research Project* — *May 2025 – Present*

- Trained a **PPO** agent in **Isaac Lab** for **last-mile manipulation**, achieving a **94% success rate**
- Developing a **hierarchical framework** for navigation and manipulation in **unknown environments using only contact sensing**

**Quadruped Path Planner for Dynamic Environments** [Website] — *Pittsburgh, PA*

*Carnegie Mellon University – Course Project* — *September 2024 – November 2024*

- Demonstrated a global path planner using C++ and ROS, that accounts for dynamic obstacles and the z-height of the robot
- Executed **Lazy-PRM with D-Star Lite** and kinodynamic constraints in Gazebo, achieving a **93% success rate**

**Model Predictive Path Integral Control** [Website] — *Pittsburgh, PA*

*Carnegie Mellon University – Course Project* — *February 2024 – April 2024*

- Implemented MPPI using C++ and ROS in simulation, with obstacle avoidance leveraging a Voxel Grid Costmap
- Benchmarked MPPI against an existing iLQR controller, demonstrating **faster path generation of up to 21%**